Computer Graphics

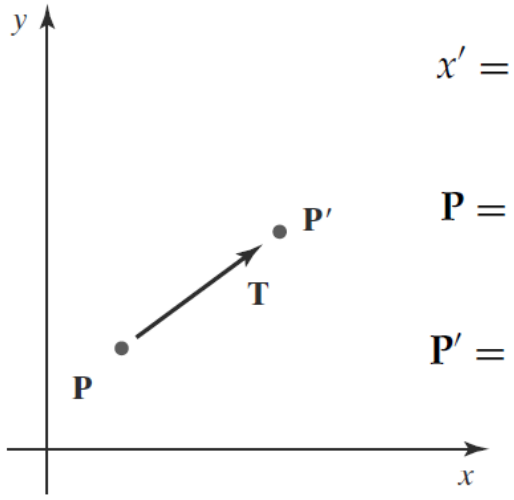
Geometric Transformations

Dongwann Kang

Associate Professor
Department of Computer Science and Engineering

Basic 2D Geometric Transformations

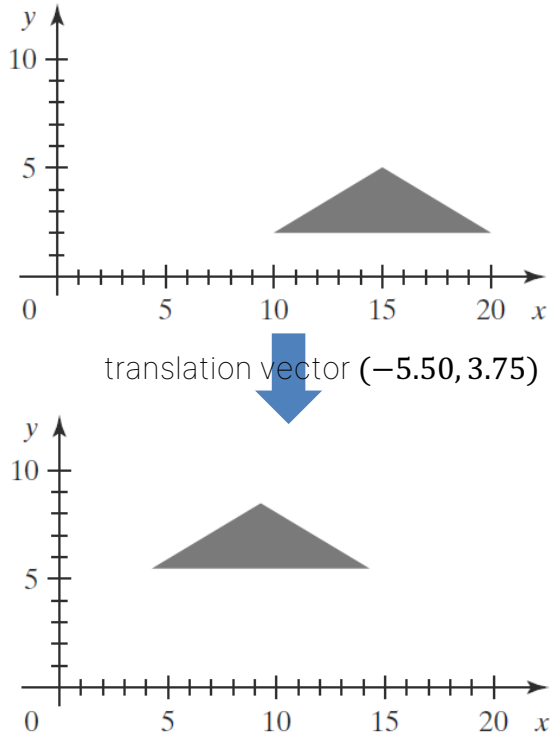
- Two-Dimensional Translation
  - A single coordinate point
    - Adding offsets to its coordinates
  - An object defined with multiple coordinate positions
    - Relocating all the coordinate positions by the same displacement along parallel paths
  - A rigid-body transformation that moves objects without deformation



$$x' = x + t_x, \qquad y' = y + t_y$$

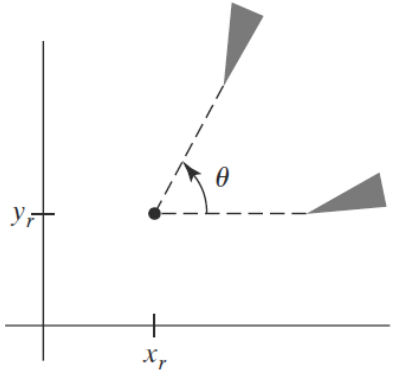
$$\mathbf{P} = \begin{bmatrix} x \\ y \end{bmatrix}, \qquad \mathbf{P}' = \begin{bmatrix} x' \\ y' \end{bmatrix}, \qquad \mathbf{T} = \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{P} + \mathbf{T}$$

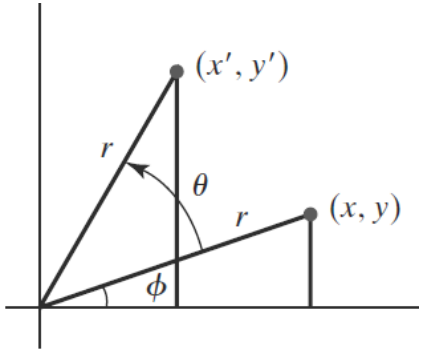


Basic 2D Geometric Transformations

- Two-Dimensional Rotation
  - Need to specify a rotation axis and a rotation angle
  - Repositioning the object along a circular path in the xy plane
    - Rotation axis is perpendicular to the xy plane
    - Pivot point: the intersection position of the rotation axis with the xy plane
    - A positive value for the angle θ defines a counterclockwise rotation about the pivot point



  - A rotation of a point position P when the pivot point is at the coordinate origin



$$x = r\cos\phi, \qquad y = r\sin\phi$$

$$x' = r\cos(\phi+\theta) = r\cos\phi\cos\theta - r\sin\phi\sin\theta$$
$$y' = r\sin(\phi+\theta) = r\cos\phi\sin\theta + r\sin\phi\cos\theta$$

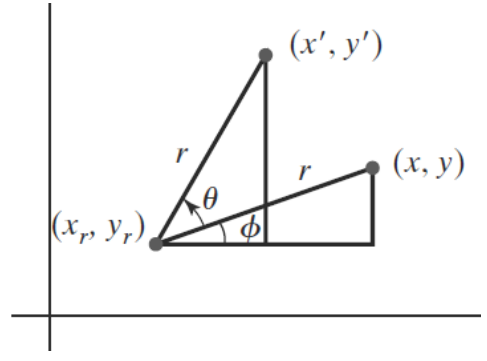
$$x' = x\cos\theta - y\sin\theta$$
$$y' = x\sin\theta + y\cos\theta$$

$$\mathbf{P}' = \mathbf{R}\cdot\mathbf{P}$$

$$\mathbf{R} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

  - Rotation of a point about an arbitrary pivot position

$$x' = x_r + (x - x_r)\cos\theta - (y - y_r)\sin\theta$$
$$y' = y_r + (x - x_r)\sin\theta + (y - y_r)\cos\theta$$



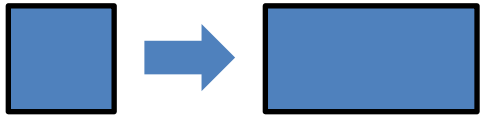
Basic 2D Geometric Transformations

- Two-Dimensional Scaling
  - Multiplying object positions (x, y) by scaling factors s_x and s_y

$$x' = x \cdot s_x, \qquad y' = y \cdot s_y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

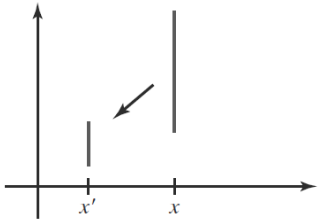
$$\mathbf{P}' = \mathbf{S} \cdot \mathbf{P}$$



scaling factors $sx = 2$ and $sy = 1$

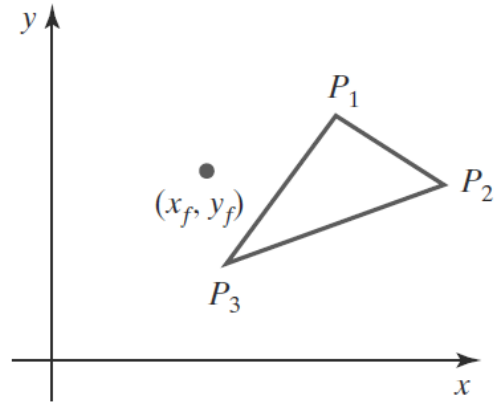
  - A uniform scaling
    - When s_x and s_y are assigned the same value, relative object proportions are maintained
  - Both scaled and repositioned
  - Scaling with the fixed point
    - The location of a scaled object remains unchanged after the scaling transformation

Scaling factors < 1



$$x' - x_f = (x - x_f)s_x, \qquad y' - y_f = (y - y_f)s_y$$

$$x' = x \cdot s_x + x_f(1 - s_x)$$
$$y' = y \cdot s_y + y_f(1 - s_y)$$



Matrix Representations and Homogeneous Coordinates

- Many graphics applications involve sequences of geometric transformations
  - Translation, rotation, and scaling can be expressed in the general matrix form
    - $\mathbf{P}' = \mathbf{M}_1 \cdot \mathbf{P} + \mathbf{M}_2$
      - $\mathbf{P}$ and $\mathbf{P}'$: coordinate positions represented as column vectors
      - $\mathbf{M}_1$: a 2 × 2 matrix containing multiplicative factors
      - $\mathbf{M}_2$: a two-element column matrix containing translational terms
      - Translation: M_1 is the identity matrix
      - Rotation and scaling: M_2 contains translational terms associated with the pivot point or scaling fixed point
    - We can obtain the final coordinate positions directly from the initial coordinates by combining the transformations!

Matrix Representations and Homogeneous Coordinates

- Homogeneous Coordinates
  - If we expand the representations to $\mathbf{3\times3}$ matrices
    - Multiplicative and translational terms for a 2D geometric transformation can be combined into a single matrix
  - To do this, expand each 2D coordinate-position representation (x, y) to a three-element representation (x_h, y_h, h), called homogeneous coordinates
    - The homogeneous parameter h is a non-zero value

$$x = \frac{x_h}{h}, \qquad y = \frac{y_h}{h}$$

      - If we simply set $h = 1$, each 2D position is represented with homogeneous coordinates $(x, y, 1)$
  - Using homogeneous coordinates allows us to represent all geometric transformation equations as matrix multiplications
    - The standard method used in graphics systems

Matrix Representations and Homogeneous Coordinates

- Two-Dimensional Translation Matrix ($\boldsymbol{T}$)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \qquad \mathbf{P}' = \mathbf{T}(t_x, t_y) \cdot \mathbf{P}$$

- Two-Dimensional Rotation Matrix ($\boldsymbol{R}$)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \qquad \mathbf{P}' = \mathbf{R}(\theta) \cdot \mathbf{P}$$

- Two-Dimensional Scaling Matrix ($\boldsymbol{S}$)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \qquad \mathbf{P}' = \mathbf{S}(s_x, s_y) \cdot \mathbf{P}$$

Inverse Transformations

- Inverse translation (T^{-1})

$$\mathbf{T}^{-1} = \begin{bmatrix} 1 & 0 & -t_x \\ 0 & 1 & -t_y \\ 0 & 0 & 1 \end{bmatrix}$$

- Inverse rotation (R^{-1})
  - Negative values for rotation angles generate rotations in a clockwise direction

$$\mathrm{R}^{-1} = \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) & 0 \\ \sin(-\theta) & \cos(-\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Transpose matrix of R

- Inverse scaling (S^{-1})

$$\mathbf{S}^{-1} = \begin{bmatrix} \frac{1}{s_x} & 0 & 0 \\ 0 & \frac{1}{s_y} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2D Composite Transformations

- We can set up a sequence of transformations as a composite transformation matrix by calculating the product of the individual transformations
  - A 2D composite transformation, $\boldsymbol{M}$, which is transforming $\boldsymbol{M_2}$ after $\boldsymbol{M_1}$

$$\begin{aligned}\mathbf{P}' &= \mathbf{M}_2 \cdot \mathbf{M}_1 \cdot \mathbf{P} \\ &= \mathbf{M} \cdot \mathbf{P}\end{aligned}$$

- Composite Two-Dimensional Translations
  - Two successive translation vectors $(\boldsymbol{t_{1x}}, \boldsymbol{t_{1y}})$ and $(\boldsymbol{t_{2x}}, \boldsymbol{t_{2y}})$

$$\begin{aligned}\mathbf{P}' &= \mathbf{T}(t_{2x}, t_{2y}) \cdot \{\mathbf{T}(t_{1x}, t_{1y}) \cdot \mathbf{P}\} \\ &= \{\mathbf{T}(t_{2x}, t_{2y}) \cdot \mathbf{T}(t_{1x}, t_{1y})\} \cdot \mathbf{P}\end{aligned}$$

$$\begin{bmatrix} 1 & 0 & t_{2x} \\ 0 & 1 & t_{2y} \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & t_{1x} \\ 0 & 1 & t_{1y} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_{1x} + t_{2x} \\ 0 & 1 & t_{1y} + t_{2y} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{T}(t_{2x}, t_{2y}) \cdot \mathbf{T}(t_{1x}, t_{1y}) = \mathbf{T}(t_{1x} + t_{2x}, t_{1y} + t_{2y})$$

2D Composite Transformations

- Composite Two-Dimensional Rotations
  - Two successive rotations

$$\begin{aligned}\mathbf{P}' &= \mathbf{R}(\theta_2) \cdot \{\mathbf{R}(\theta_1) \cdot \mathbf{P}\} \\ &= \{\mathbf{R}(\theta_2) \cdot \mathbf{R}(\theta_1)\} \cdot \mathbf{P}\end{aligned}$$

$$\mathbf{R}(\theta_2) \cdot \mathbf{R}(\theta_1) = \mathbf{R}(\theta_1 + \theta_2)$$

$$\mathbf{P}' = \mathbf{R}(\theta_1 + \theta_2) \cdot \mathbf{P}$$

- Composite Two-Dimensional Scalings
  - Two successive scaling operations

$$\begin{bmatrix} s_{2x} & 0 & 0 \\ 0 & s_{2y} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s_{1x} & 0 & 0 \\ 0 & s_{1y} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} s_{1x} \cdot s_{2x} & 0 & 0 \\ 0 & s_{1y} \cdot s_{2y} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

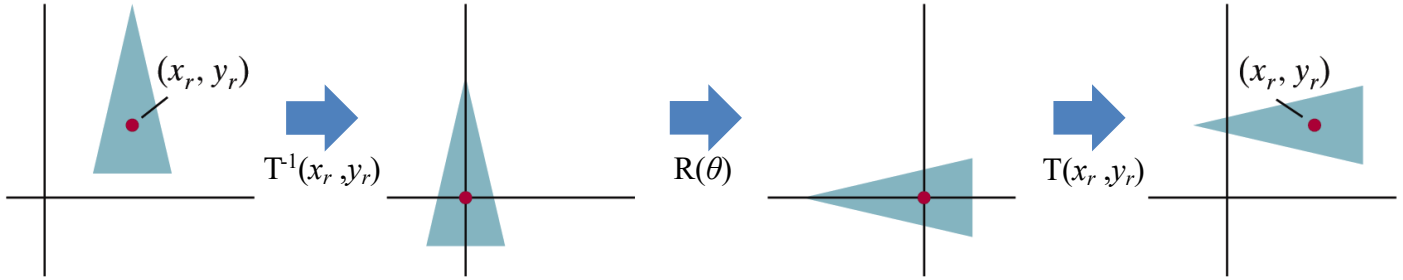
$$\mathbf{S}(s_{2x}, s_{2y}) \cdot \mathbf{S}(s_{1x}, s_{1y}) = \mathbf{S}(s_{1x} \cdot s_{2x}, \quad s_{1y} \cdot s_{2y})$$

2D Composite Transformations

- General Two-Dimensional Pivot-Point Rotation
  1. Translate the object so that the pivot-point position is moved to the coordinate origin
  2. Rotate the object about the coordinate origin
  3. Translate the object so that the pivot point is returned to its original position

$$\begin{bmatrix} 1 & 0 & x_r \\ 0 & 1 & y_r \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -x_r \\ 0 & 1 & -y_r \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & x_r(1-\cos\theta) + y_r \sin\theta \\ \sin\theta & \cos\theta & y_r(1-\cos\theta) - x_r \sin\theta \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{T}(x_r, y_r) \cdot \mathbf{R}(\theta) \cdot \underbrace{\mathbf{T}(-x_r, -y_r)}_{\mathbf{T}^{-1}(x_r, y_r)} = \mathbf{R}(x_r, y_r, \theta)$$

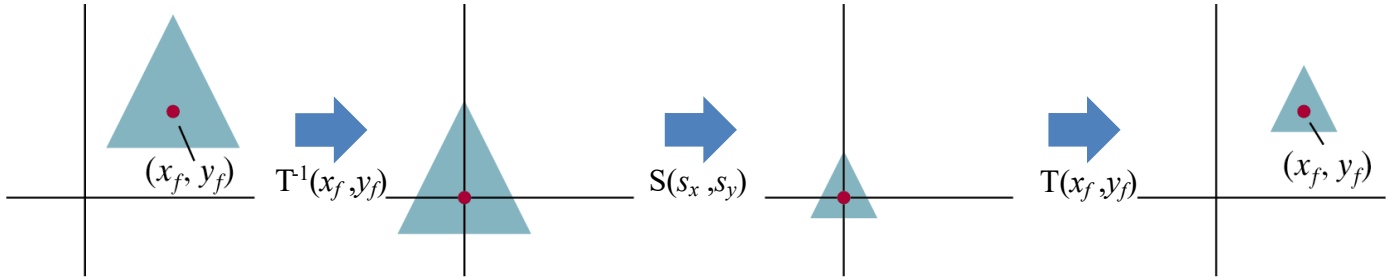


2D Composite Transformations

- General Two-Dimensional Fixed-Point Scaling
  1. Translate the object so that the fixed point coincides with the coordinate origin
  2. Scale the object with respect to the coordinate origin
  3. Use the inverse of the translation in step (1) to return the object to its original position

$$\begin{bmatrix} 1 & 0 & x_f \\ 0 & 1 & y_f \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -x_f \\ 0 & 1 & -y_f \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & x_f(1-s_x) \\ 0 & s_y & y_f(1-s_y) \\ 0 & 0 & 1 \end{bmatrix}$$

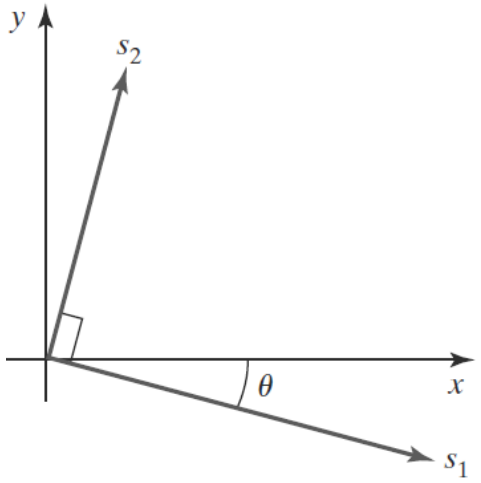
$$\mathbf{T}(x_f, y_f) \cdot \mathbf{S}(s_x, s_y) \cdot \underbrace{\mathbf{T}(-x_f, -y_f)}_{\mathbf{T}^{-1}(x_r, y_r)} = \mathbf{S}(x_f, y_f, s_x, s_y)$$

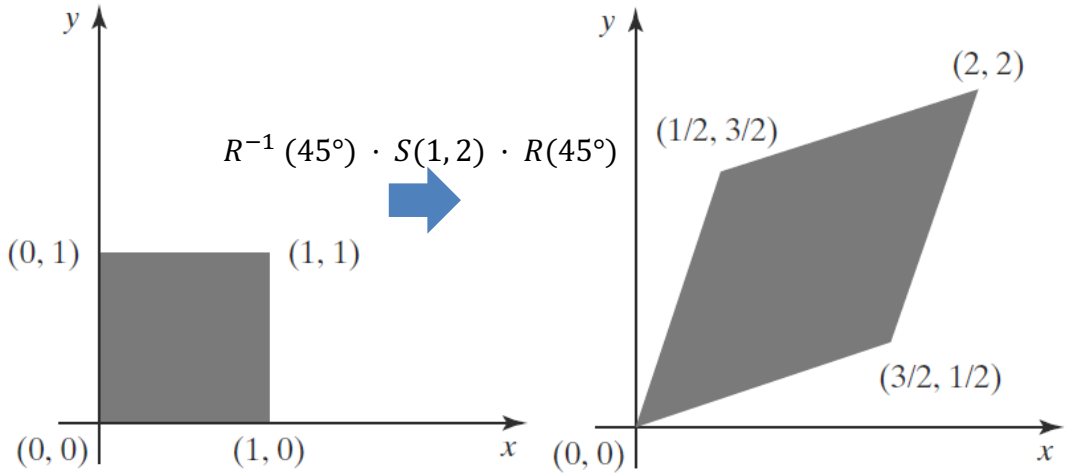


2D Composite Transformations

- General Two-Dimensional Scaling Directions
  - Parameters s_x and s_y scale objects along the x and y directions
  - We can scale an object in other directions
    1. Perform a rotation so that the directions for s_1 and s_2 coincide with the x and y axes, respectively
    2. Apply the scaling transformation $S(s_1, s_2)$
    3. Perform an opposite rotation to return points to their original orientations

$$\mathbf{R}^{-1}(\theta) \cdot \mathbf{S}(s_1, s_2) \cdot \mathbf{R}(\theta) = \begin{bmatrix} s_1 \cos^2\theta + s_2 \sin^2\theta & (s_2 - s_1)\cos\theta \sin\theta & 0 \\ (s_2 - s_1)\cos\theta \sin\theta & s_1 \sin^2\theta + s_2 \cos^2\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$





2D Composite Transformations

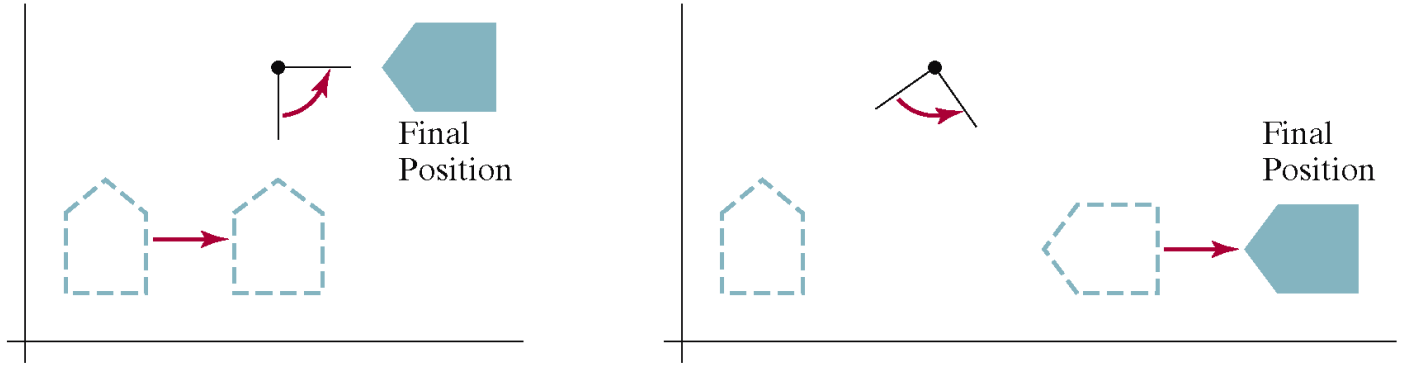
- Matrix Concatenation Properties
  - Multiplication of matrices is associative

$$\mathbf{M}_3 \cdot \mathbf{M}_2 \cdot \mathbf{M}_1 = (\mathbf{M}_3 \cdot \mathbf{M}_2) \cdot \mathbf{M}_1 = \mathbf{M}_3 \cdot (\mathbf{M}_2 \cdot \mathbf{M}_1)$$

  - On the other hand, transformation products may not be commutative

$$\mathbf{M}_2 \cdot \mathbf{M}_1 \neq \mathbf{M}_1 \cdot \mathbf{M}_2$$

    - For some special cases, the multiplication of transformation matrices is commutative
      - A sequence of transformations that are all of the same kind
        - » Two successive rotations
      - Rotation and uniform scaling



2D Composite Transformations

- General 2D Composite Transformations and Computational Efficiency
  - A 2D transformation representing any combination of translations, rotations, and scalings

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} rs_{xx} & rs_{xy} & trs_x \\ rs_{yx} & rs_{yy} & trs_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

only four multiplications and four additions

$$x' = x \cdot rs_{xx} + y \cdot rs_{xy} + trs_x, \qquad y' = x \cdot rs_{yx} + y \cdot rs_{yy} + trs_y$$

- E.g.) an object scaled and rotated about its centroid coordinates (x_c, y_c) and then translated

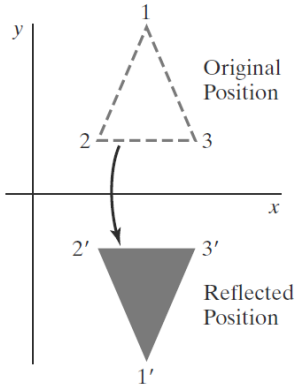
$$\mathbf{T}(t_x, t_y) \cdot \mathbf{R}(x_c, y_c, \theta) \cdot \mathbf{S}(x_c, y_c, s_x, s_y)$$

$$= \begin{bmatrix} s_x \cos\theta & -s_y \sin\theta & x_c(1 - s_x \cos\theta) + y_c s_y \sin\theta + t_x \\ s_x \sin\theta & s_y \cos\theta & y_c(1 - s_y \cos\theta) - x_c s_x \sin\theta + t_y \\ 0 & 0 & 1 \end{bmatrix}$$

Other Two-Dimensional Transformations

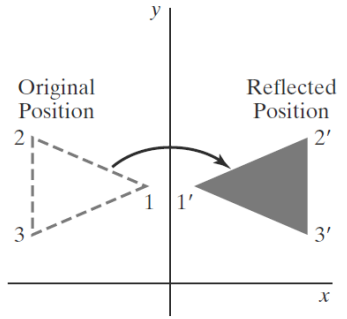
- Reflection
  - Reflection about the line $y = 0$ (the x axis)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



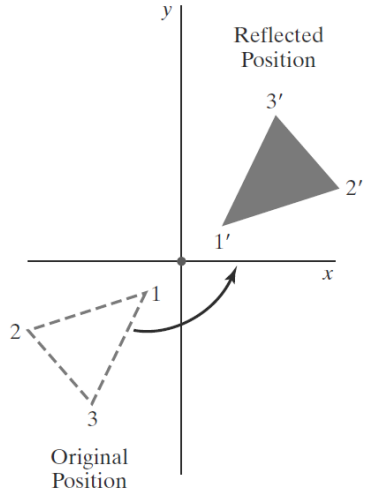
  - Reflection about the line $x = 0$ (the y axis)

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



  - Reflection to the coordinate origin
    - The same as the rotation matrix $R(\theta)$ with $\theta = 180°$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



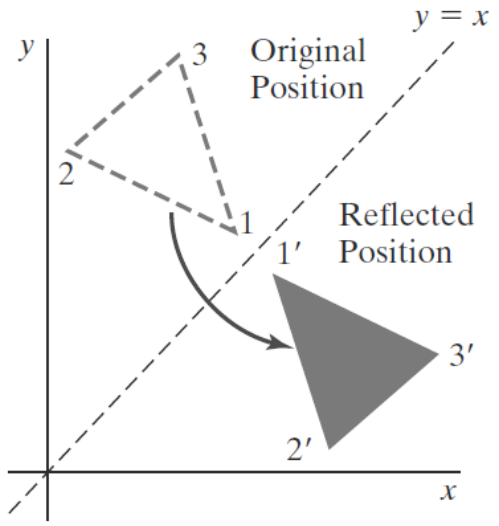
Other Two-Dimensional Transformations

- Reflection
  - Reflection about the diagonal line $y = x$
    - A clockwise rotation by 45°
    - A reflection about the x axis
    - A conterclockwise rotation by 45°

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

  - Reflection about the diagonal line $y = -x$
    - A clockwise rotation by 45°
    - A reflection about the y axis
    - A conterclockwise rotation by 45°

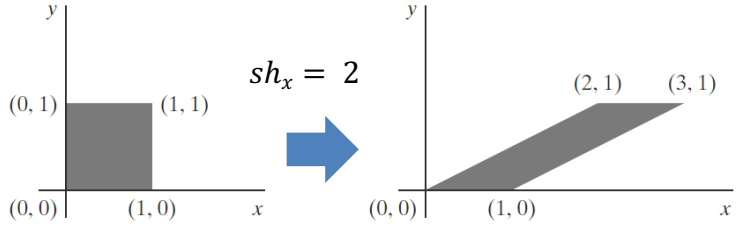
$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Other Two-Dimensional Transformations

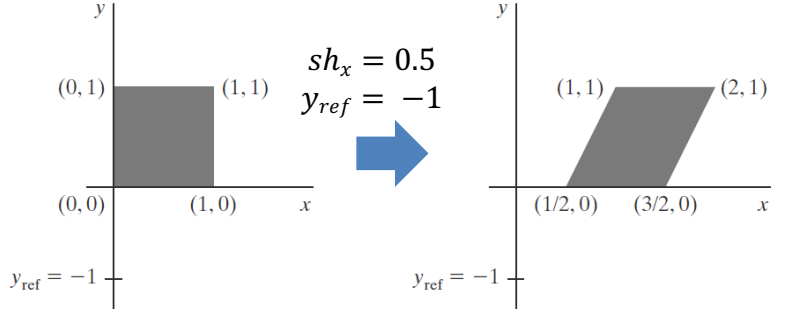
- Shear
  - An $\boldsymbol{x}$-direction shear

$$\begin{bmatrix} 1 & sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad x' = x + sh_x \cdot y, \qquad y' = y$$



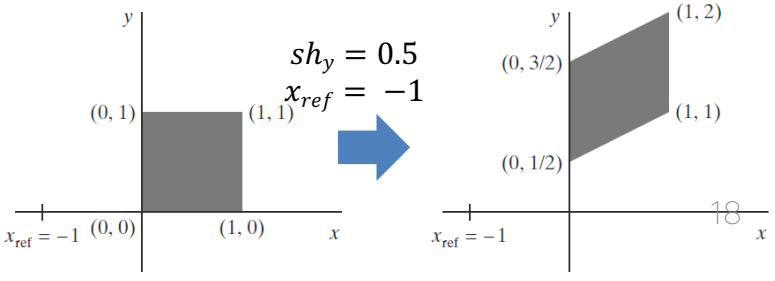
  - An $\boldsymbol{x}$-direction shear relative to the line $\boldsymbol{y = y_{ref}}$

$$\begin{bmatrix} 1 & sh_x & -sh_x \cdot y_{\text{ref}} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad x' = x + sh_x(y - y_{\text{ref}}), \qquad y' = y$$



  - A $\boldsymbol{y}$-direction shear relative to the line $\boldsymbol{x = x_{ref}}$

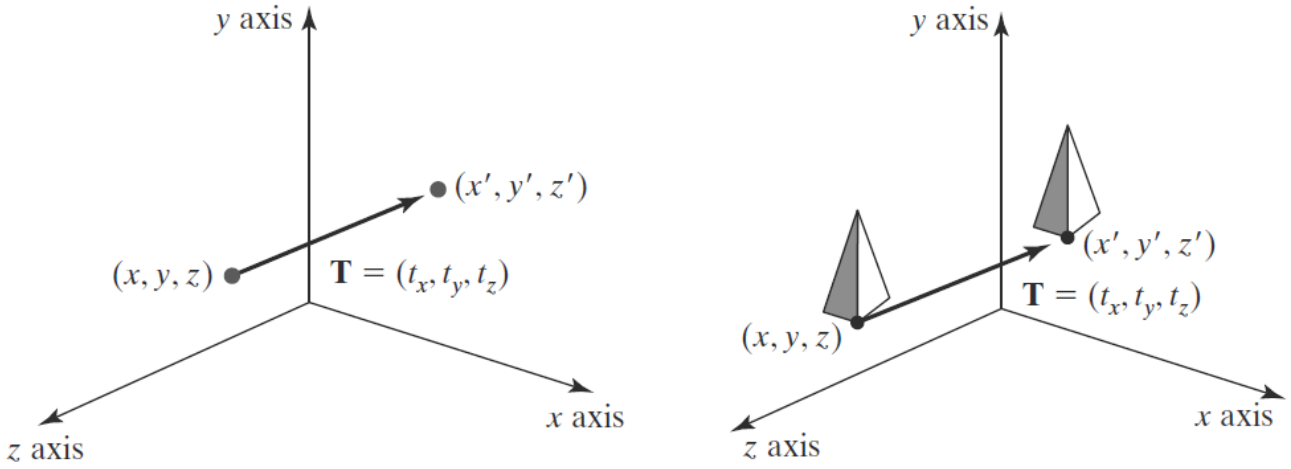
$$\begin{bmatrix} 1 & 0 & 0 \\ sh_y & 1 & -sh_y \cdot x_{\text{ref}} \\ 0 & 0 & 1 \end{bmatrix} \qquad x' = x, \qquad y' = y + sh_y(x - x_{\text{ref}})$$



Three-Dimensional Translation

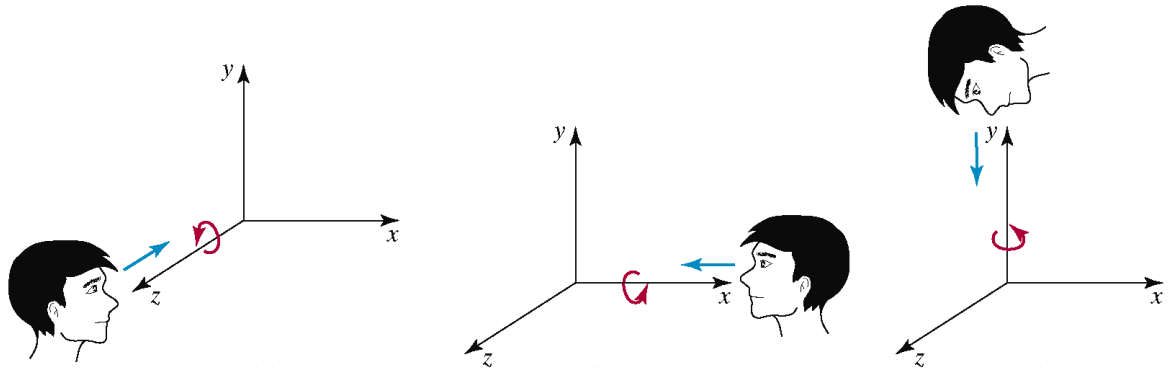
$$x' = x + t_x, \qquad y' = y + t_y, \qquad z' = z + t_z$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \qquad \mathbf{P}' = \mathbf{T} \cdot \mathbf{P}$$



Three-Dimensional Rotation

- Positive rotations about a coordinate axis are counterclockwise, when looking along the positive half of the axis toward the origin

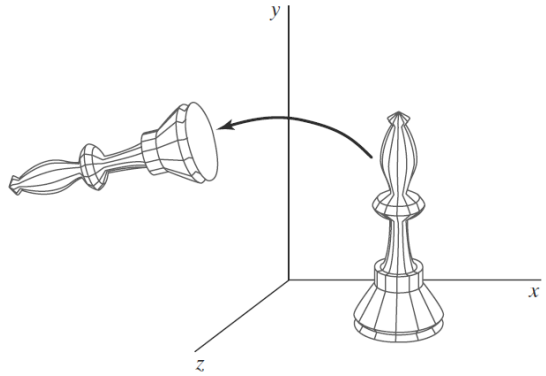


- Three-Dimensional Coordinate-Axis Rotations
  - The two-dimensional z-axis rotation

$$x' = x\cos\theta - y\sin\theta$$
$$y' = x\sin\theta + y\cos\theta$$
$$z' = z$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{R}_z(\theta) \cdot \mathbf{P}$$



Three-Dimensional Rotation

- Three-Dimensional Coordinate-Axis Rotations
  - The two-dimensional x-axis rotation

$$y' = y\cos\theta - z\sin\theta$$
$$z' = y\sin\theta + z\cos\theta$$
$$x' = x$$

  - The two-dimensional y-axis rotation

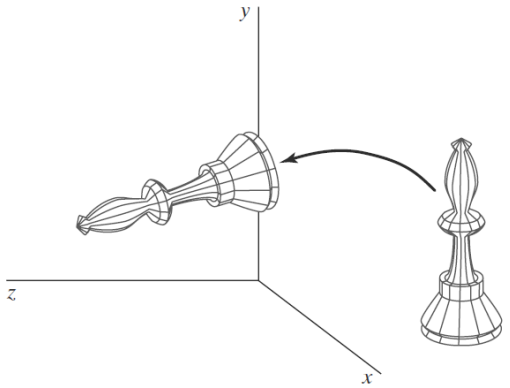
$$z' = z\cos\theta - x\sin\theta$$
$$x' = z\sin\theta + x\cos\theta$$
$$y' = y$$

  - Inverse rotation
    - Transpose matrix

The two-dimensional z-axis rotation

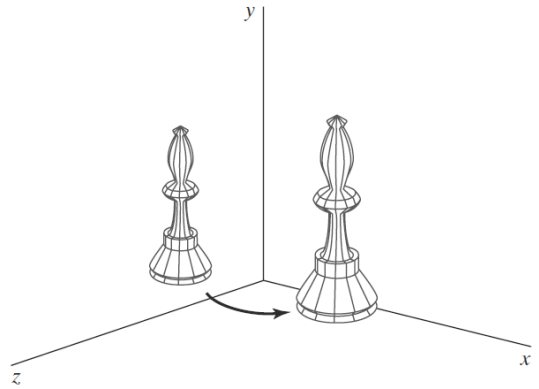
$$x' = x\cos\theta - y\sin\theta$$
$$y' = x\sin\theta + y\cos\theta$$
$$z' = z$$

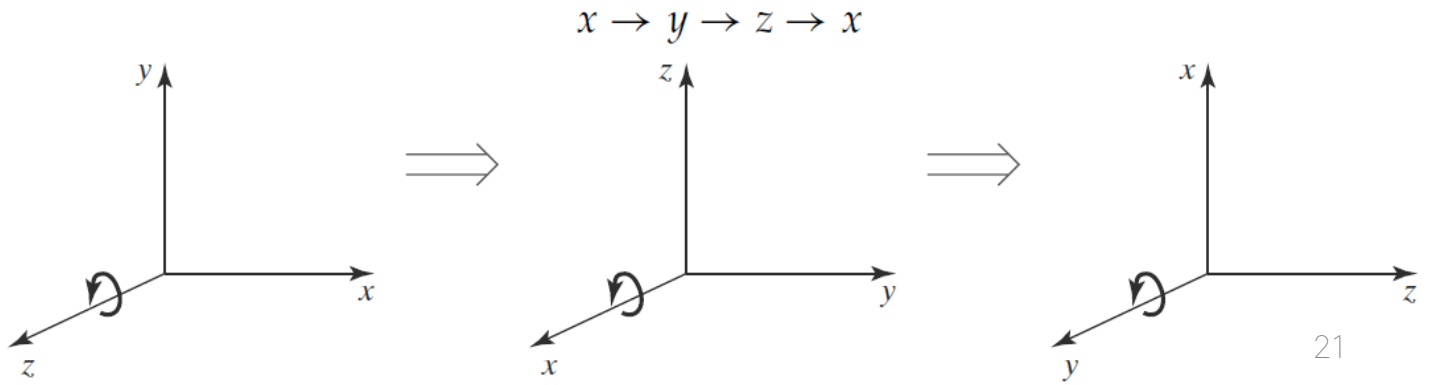
$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

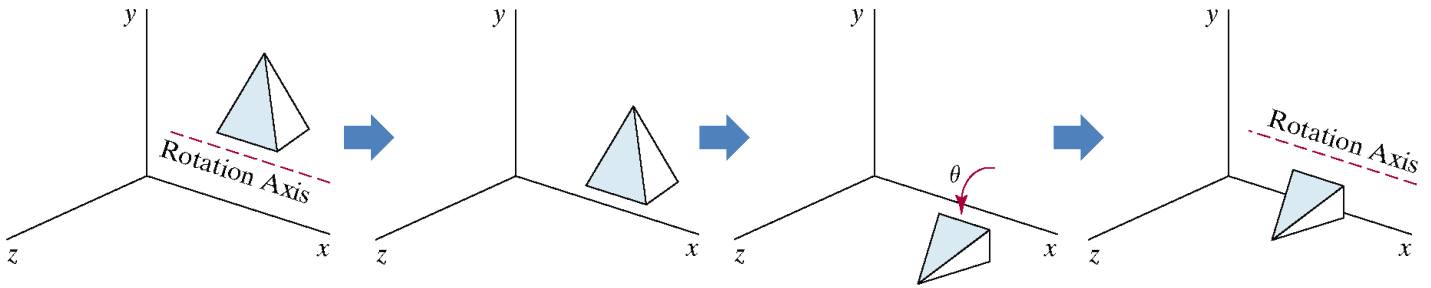




Three-Dimensional Rotation

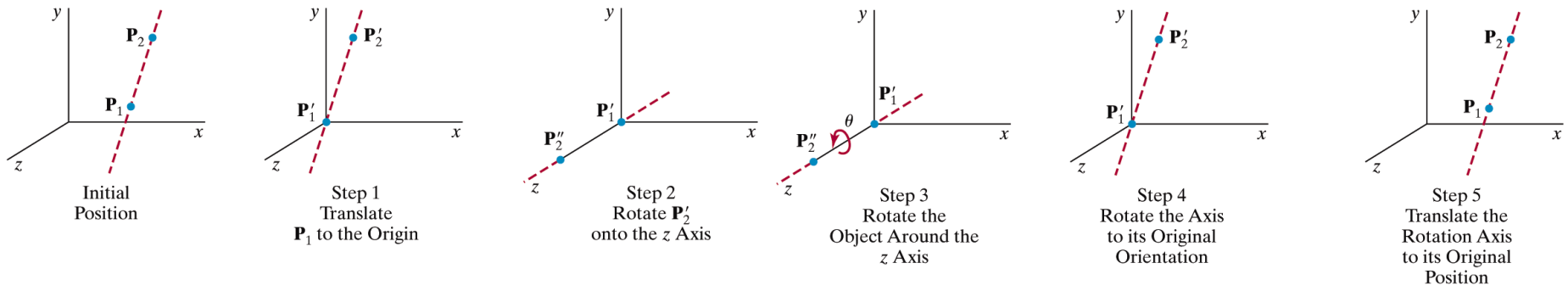
- General Three-Dimensional Rotations
  - A rotation matrix for any axis that does not coincide with a coordinate axis
    - A combinations of translations and the coordinate-axis rotations
  - An object that is rotated about an axis that is parallel to one of the coordinate axes
    1. Translate the object so that the rotation axis coincides with the parallel coordinate axis
    2. Perform the specified rotation about that axis
    3. Translate the object so that the rotation axis is moved back to its original position

$$\mathbf{P}' = \mathbf{T}^{-1} \cdot \mathbf{R}_x(\theta) \cdot \mathbf{T} \cdot \mathbf{P}$$
$$\mathbf{R}(\theta) = \mathbf{T}^{-1} \cdot \mathbf{R}_x(\theta) \cdot \mathbf{T}$$



Three-Dimensional Rotation

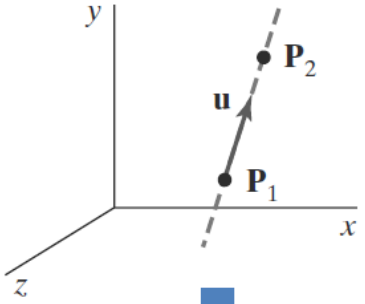
- General Three-Dimensional Rotations
  - An object that is rotated about an axis that is not parallel to one of the coordinate axes
    1. Translate the object so that the rotation axis passes through the coordinate origin
    2. Rotate the object so that the axis of rotation coincides with one of the coordinate axes
    3. Perform the specified rotation about the selected coordinate axis
    4. Apply inverse rotations to bring the rotation axis back to its original orientation
    5. Apply the inverse translation to bring the rotation axis back to its original spatial position



Three-Dimensional Rotation

- General Three-Dimensional Rotations
  - An object that is rotated about an axis that is not parallel to one of the coordinate axes
    - Assume that the rotation axis is defined by two points, and that the direction of rotation is counterclockwise when looking along the axis from P_2 to P_1
    - The rotation-axis vector V

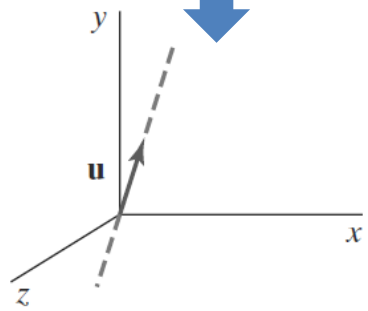
$$\mathbf{V} = \mathbf{P}_2 - \mathbf{P}_1 = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$



    - The unit rotation-axis vector u

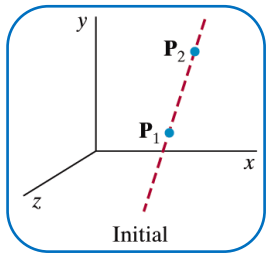
$$\mathbf{u} = \frac{\mathbf{V}}{|\mathbf{V}|} = (a, b, c)$$

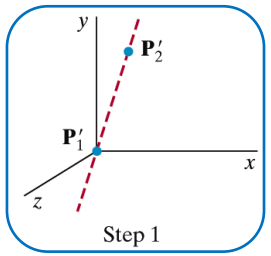
$$a = \frac{x_2 - x_1}{|\mathbf{V}|}, \qquad b = \frac{y_2 - y_1}{|\mathbf{V}|}, \qquad c = \frac{z_2 - z_1}{|\mathbf{V}|}$$

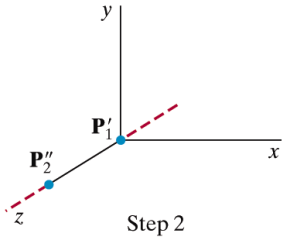


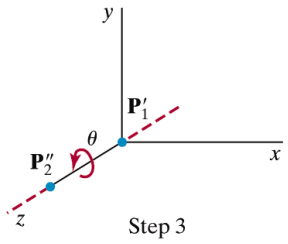
    - Move the point P_1 to the origin

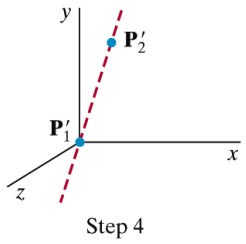
$$\mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & -x_1 \\ 0 & 1 & 0 & -y_1 \\ 0 & 0 & 1 & -z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

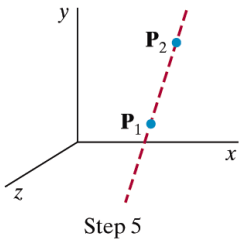






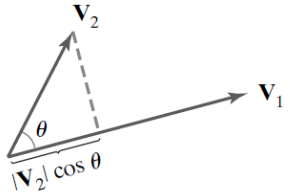




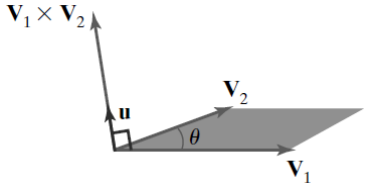


Three-Dimensional Rotation

- General Three-Dimensional Rotations
  - An object that is rotated about an axis that is not parallel to one of the coordinate axes
    - Formulate the transformations that will put the rotation axis onto the z axis
      - Rotate about the x axis
        - » Gets vector u into the xz plane
      - Then rotate about the y axis
        - » swings u around to the z axis
      - The projection of u in the yz plane as the vector $u' = (0, b, c)$



$$\mathbf{V}_1 \cdot \mathbf{V}_2 = |\mathbf{V}_1|\,|\mathbf{V}_2| \cos\theta, \qquad 0 \le \theta \le \pi$$



$$\mathbf{V}_1 \times \mathbf{V}_2 = \mathbf{u}|\mathbf{V}_1|\,|\mathbf{V}_2| \sin\theta, \qquad 0 \le \theta \le \pi$$

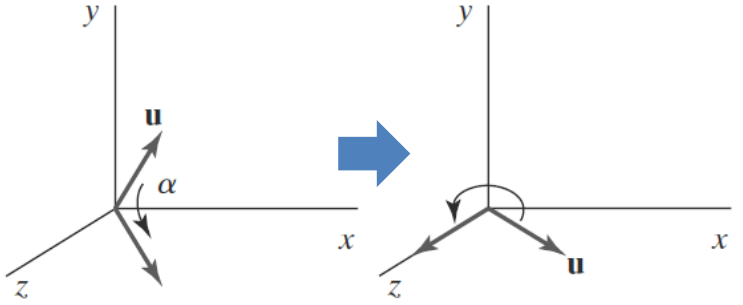
$$\mathbf{V}_1 \times \mathbf{V}_2 = (V_{1y}V_{2z} - V_{1z}V_{2y},\ V_{1z}V_{2x} - V_{1x}V_{2z},\ V_{1x}V_{2y} - V_{1y}V_{2x})$$

$$\cos\alpha = \frac{\mathbf{u}' \cdot \mathbf{u}_z}{|\mathbf{u}'|\,|\mathbf{u}_z|} = \frac{c}{d} \qquad d = \sqrt{b^2 + c^2}$$

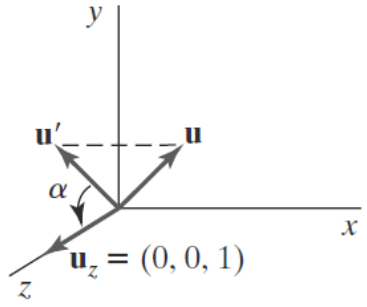
$$\mathbf{u}' \times \mathbf{u}_z = \mathbf{u}_x\,|\mathbf{u}'|\,|\mathbf{u}_z| \sin\alpha$$

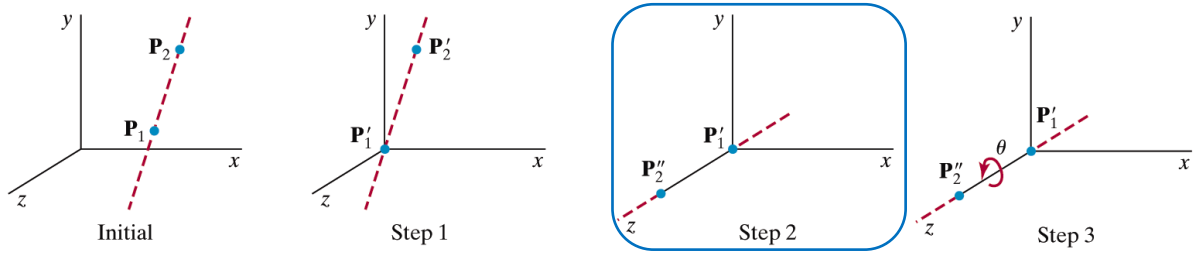
$$\mathbf{u}' \times \mathbf{u}_z = \mathbf{u}_x \cdot b$$

$$d \sin\alpha = b \qquad \sin\alpha = \frac{b}{d}$$



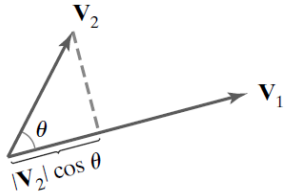
$$\mathbf{R}_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{c}{d} & -\frac{b}{d} & 0 \\ 0 & \frac{b}{d} & \frac{c}{d} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

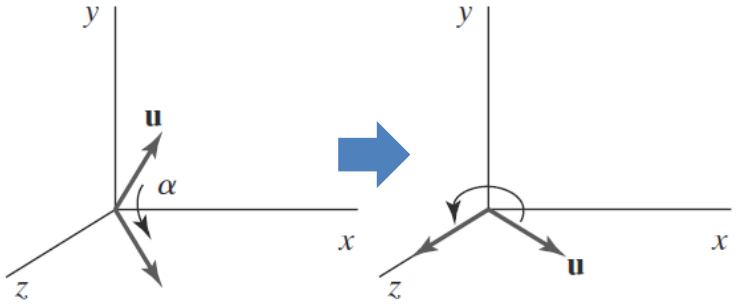




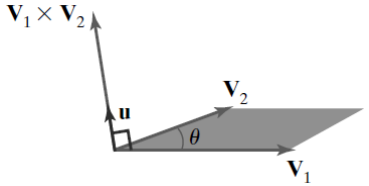
Three-Dimensional Rotation

- General Three-Dimensional Rotations
  - An object that is rotated about an axis that is not parallel to one of the coordinate axes
    - Formulate the transformations that will put the rotation axis onto the z axis
      - Rotate about the x axis
        - » Gets vector u into the xz plane
      - Then rotate about the y axis
        - » swings u around to the z axis
      - The rotated vector $u'' = (a, 0, d)$





$$\mathbf{V}_1 \cdot \mathbf{V}_2 = |\mathbf{V}_1|\,|\mathbf{V}_2| \cos\theta, \qquad 0 \le \theta \le \pi$$



$$\mathbf{V}_1 \times \mathbf{V}_2 = \mathbf{u}|\mathbf{V}_1|\,|\mathbf{V}_2| \sin\theta, \qquad 0 \le \theta \le \pi$$

$$\mathbf{V}_1 \times \mathbf{V}_2 = (V_{1y}V_{2z} - V_{1z}V_{2y},\ V_{1z}V_{2x} - V_{1x}V_{2z},\ V_{1x}V_{2y} - V_{1y}V_{2x})$$

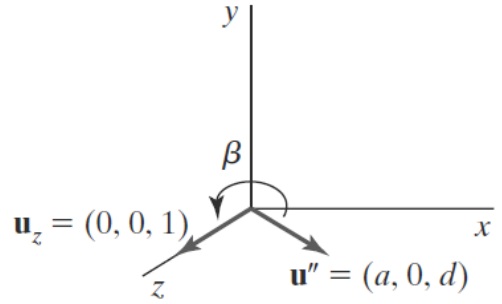
$$\cos\beta = \frac{\mathbf{u}'' \cdot \mathbf{u}_z}{|\mathbf{u}''|\,|\mathbf{u}_z|} = d$$

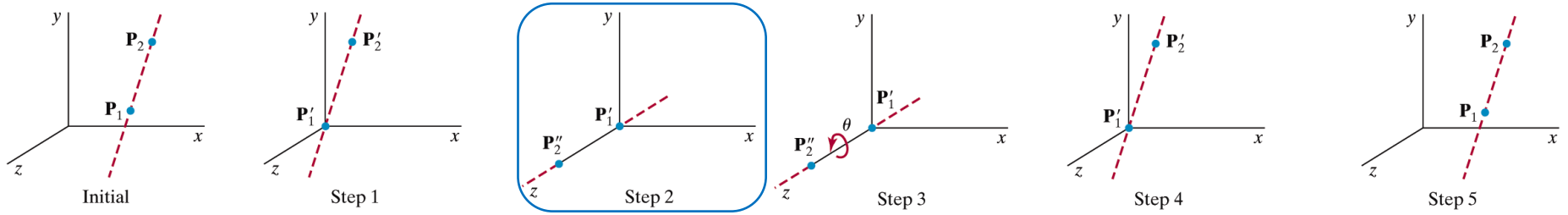
$$\mathbf{u}'' \times \mathbf{u}_z = \mathbf{u}_y\,|\mathbf{u}''|\,|\mathbf{u}_z| \sin\beta$$

$$\mathbf{u}'' \times \mathbf{u}_z = \mathbf{u}_y \cdot (-a)$$

$$\sin\beta = -a$$

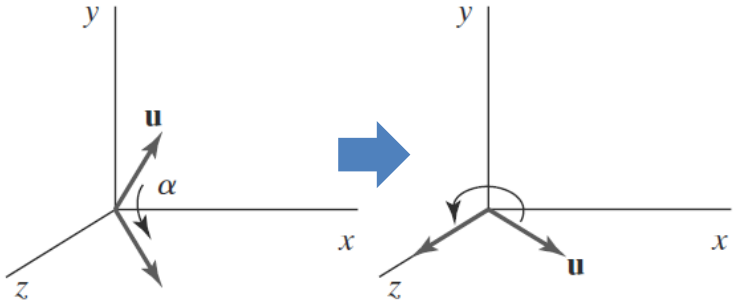
$$\mathbf{R}_y(\beta) = \begin{bmatrix} d & 0 & -a & 0 \\ 0 & 1 & 0 & 0 \\ a & 0 & d & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



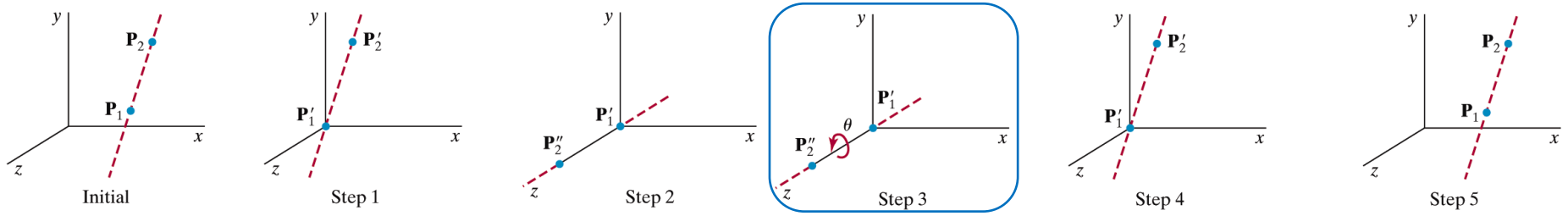


Three-Dimensional Rotation

- General Three-Dimensional Rotations
  - An object that is rotated about an axis that is not parallel to one of the coordinate axes
    - Formulate the transformations that will put the rotation axis onto the z axis
      - Rotate about the x axis
        - » Gets vector u into the xz plane
      - Then rotate about the y axis
        - » swings u around to the z axis
    - Rotation about the z axis by $\theta°$

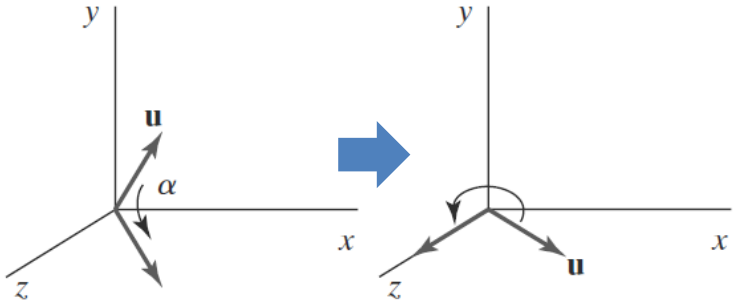


$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{R}_y(\beta) = \begin{bmatrix} d & 0 & -a & 0 \\ 0 & 1 & 0 & 0 \\ a & 0 & d & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{R}_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{c}{d} & -\frac{b}{d} & 0 \\ 0 & \frac{b}{d} & \frac{c}{d} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & -x_1 \\ 0 & 1 & 0 & -y_1 \\ 0 & 0 & 1 & -z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



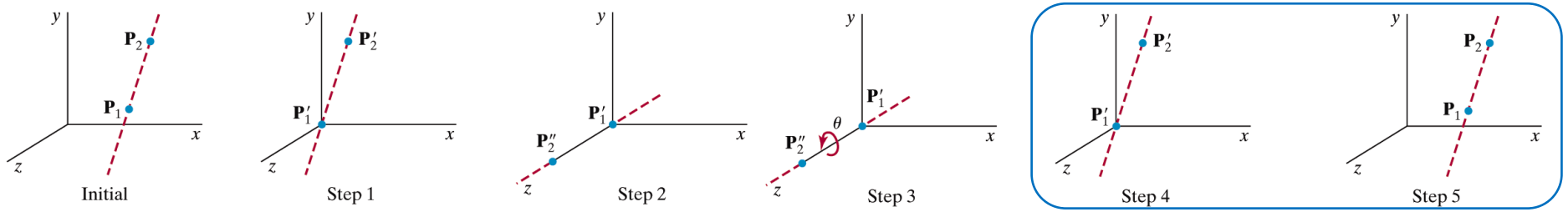
Three-Dimensional Rotation

- General Three-Dimensional Rotations
  - An object that is rotated about an axis that is not parallel to one of the coordinate axes
    - Formulate the transformations that will put the rotation axis onto the z axis
      - Rotate about the x axis
        - » Gets vector u into the xz plane
      - Then rotate about the y axis
        - » swings u around to the z axis
    - Rotation about the z axis by $\theta°$
    - Transform the rotation axis back to its original position



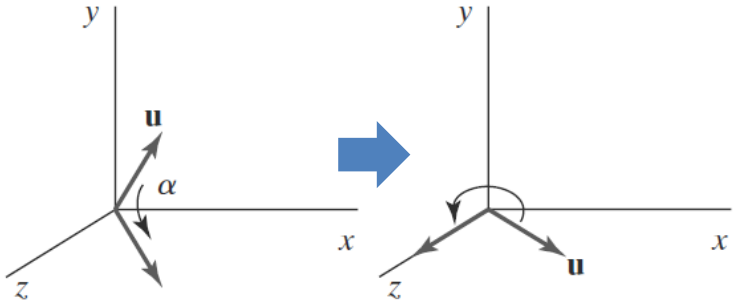
$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{R}_y(\beta) = \begin{bmatrix} d & 0 & -a & 0 \\ 0 & 1 & 0 & 0 \\ a & 0 & d & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{R}_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{c}{d} & -\frac{b}{d} & 0 \\ 0 & \frac{b}{d} & \frac{c}{d} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & -x_1 \\ 0 & 1 & 0 & -y_1 \\ 0 & 0 & 1 & -z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}(\theta) = \mathbf{T}^{-1} \cdot \mathbf{R}_x^{-1}(\alpha) \cdot \mathbf{R}_y^{-1}(\beta) \cdot \mathbf{R}_z(\theta) \cdot \mathbf{R}_y(\beta) \cdot \mathbf{R}_x(\alpha) \cdot \mathbf{T}$$



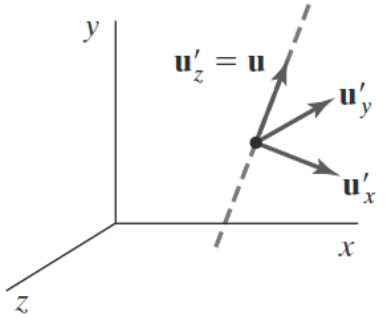
Three-Dimensional Rotation

- General Three-Dimensional Rotations
  - An object that is rotated about an axis that is not parallel to one of the coordinate axes
    - Formulate the transformations that will put the rotation axis onto the z axis
      - Rotate about the x axis
        - » Gets vector u into the xz plane
      - Then rotate about the y axis
        - » swings u around to the z axis
    - Rotation about the z axis by $\theta°$
    - Transform the rotation axis back to its original position



$$\mathbf{R}_y(\beta) = \begin{bmatrix} d & 0 & -a & 0 \\ 0 & 1 & 0 & 0 \\ a & 0 & d & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{R}_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{c}{d} & -\frac{b}{d} & 0 \\ 0 & \frac{b}{d} & \frac{c}{d} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- The matrix R transforms the unit vectors u'_x, u'_y, and u'_z onto the x, y, and z axes, respectively



$$\mathbf{R} = \begin{bmatrix} u'_{x1} & u'_{x2} & u'_{x3} & 0 \\ u'_{y1} & u'_{y2} & u'_{y3} & 0 \\ u'_{z1} & u'_{z2} & u'_{z3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

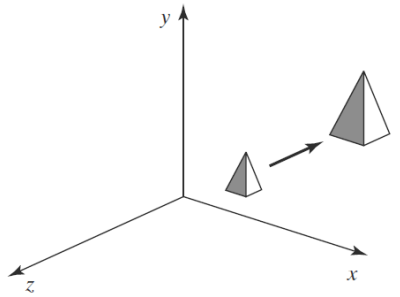
$$\mathbf{u}'_x = (u'_{x1}, u'_{x2}, u'_{x3})$$
$$\mathbf{u}'_y = (u'_{y1}, u'_{y2}, u'_{y3})$$
$$\mathbf{u}'_z = (u'_{z1}, u'_{z2}, u'_{z3})$$

Three-Dimensional Scaling

- The three-dimensional scaling transformation of a position $P = (x, y, z)$ relative to the coordinate origin

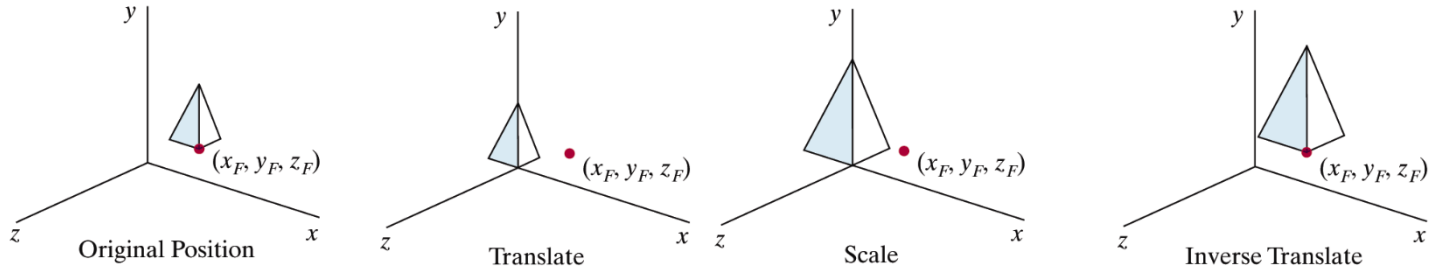
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \qquad \mathbf{P}' = \mathbf{S} \cdot \mathbf{P}$$

$$x' = x \cdot s_x, \qquad y' = y \cdot s_y, \qquad z' = z \cdot s_z$$



- A scaling transformation with respect to any selected fixed position (x_f, y_f, z_f)

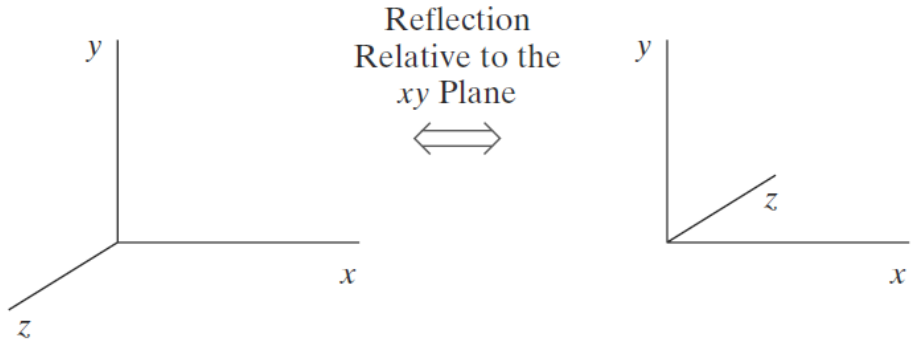
$$\mathbf{T}(x_f, y_f, z_f) \cdot \mathbf{S}(s_x, s_y, s_z) \cdot \mathbf{T}(-x_f, -y_f, -z_f) = \begin{bmatrix} s_x & 0 & 0 & (1 - s_x)x_f \\ 0 & s_y & 0 & (1 - s_y)y_f \\ 0 & 0 & s_z & (1 - s_z)z_f \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Other Three-Dimensional Transformations

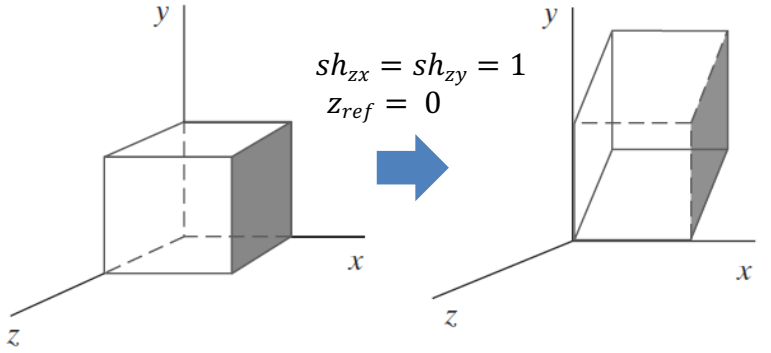
- Three-Dimensional Reflections
  - An example of a reflection that converts coordinate specifications from a right-handed system to a left-handed system

$$M_{\text{zreflect}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



- Three-Dimensional Shears
  - A general z-axis shearing transformation

$$M_{\text{zshear}} = \begin{bmatrix} 1 & 0 & \text{sh}_{zx} & -\text{sh}_{zx} \cdot z_{\text{ref}} \\ 0 & 1 & \text{sh}_{zy} & -\text{sh}_{zy} \cdot z_{\text{ref}} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Affine Transformations

- Affine transformation

$$x' = a_{xx}x + a_{xy}y + a_{xz}z + b_x$$
$$y' = a_{yx}x + a_{yy}y + a_{yz}z + b_y$$
$$z' = a_{zx}x + a_{zy}y + a_{zz}z + b_z$$

  - Each of the transformed coordinates $\boldsymbol{x'}$, $\boldsymbol{y'}$, and $\boldsymbol{z'}$ is a linear function of the original coordinates $\boldsymbol{x}$, $\boldsymbol{y}$, and $\boldsymbol{z}$
  - Parameters $\boldsymbol{a_{ij}}$ and $\boldsymbol{b_k}$ are constants determined by the transformation type

  - Parallel lines are transformed into parallel lines, and finite points map to finite points
    - Translation, rotation, scaling, reflection, and shear
    - The conversion of coordinates for a scene from one reference system to another
    - An affine transformation involving only translation, rotation, and reflection
      - Preserves angles and lengths, as well as parallel lines